

Solution Requirements

Date	26 March 2026
Project Name	Smart Lender

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system shall allow Admin and Users to securely log in using a username and password before accessing the application.	High
2	Authorization Levels	The admin can manage users, datasets, machine learning models, and prediction records. Users can register, log in, submit loan applications, and view prediction results.	High
3	External Interfaces	The system shall interact with the machine learning prediction module, local database, and web browser interface for processing loan	High

		applications and displaying results.	
4	Transactions Processing	The system shall process loan application data, validate user inputs, generate loan approval predictions, and store prediction history in the database.	High
5	Reporting	The system shall generate prediction reports, applicant history, model performance summaries, and downloadable result records for the administrator.	Medium
6	Business Rules	Loan approval decisions shall be based on applicant details such as income, credit history, loan amount, employment status, education, and other predefined eligibility criteria using trained machine learning models.	High
7	Compliance to Laws or Regulations	The system shall ensure secure handling of user information by following applicable data privacy and security guidelines and protecting sensitive applicant data.	Medium
8	Other	The system shall provide an easy-to-use interface for loan prediction and maintain prediction history for future reference.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system shall quickly process loan applications and display prediction results with minimal delay.	Prediction results generated within 2 seconds.
2	Scalability	The system shall support increasing numbers of users and loan prediction requests without significant performance degradation.	Supports up to 1,000 concurrent users.
3	Security & Data Privacy	The system shall protect applicant information using secure authentication, encrypted data storage, and restricted access.	Password encryption and secure user authentication with role-based access control.
4	Reliability & Availability	The application shall remain operational with minimal downtime and ensure accurate prediction of results.	99% system availability with reliable prediction processing.
5	Usability & Accessibility	The system shall provide a simple, responsive, and user-friendly interface that is easy to navigate across different devices.	Users should complete a loan prediction in less than 3 minutes without assistance.
6	Other	The system shall be easy to maintain, update, and extend with new machine learning models or additional features in the future.	Modular architecture with well-documented code and database structure.

